

Hitkumar (Heet) Patel

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TECHNICAL PROJECTS

Qibo (Open Source Contribution) | Python, Quantum Computing

- Contributed to the **Qibo** quantum simulation framework by implementing a **Fast Walsh-Hadamard Transform (FHT)** for Pauli-Liouville basis conversion.
- Optimized computational complexity to $O(4^n \cdot n)$, improving performance for high-dimensional quantum state simulations.
- Collaborated with maintainers via GitHub PRs to ensure code compliance with open-source framework standards.

Situate Vancouver | React, MapLibre, Django, FastAPI

- Working with a team to engineer a "city intelligence" workspace for Metro Vancouver, providing an interactive platform for exploring mobility and transit data.
- Solved performance bottlenecks by managing heavy map interaction layers and camera updates, preventing UI lag during real-time data filtering.
- Set up local environments for iOS and android QA testing.
- Live at: situatevancouver.com

Spatial Audio Engine | Web Audio API, JavaScript, Canvas

- Developed an interactive 3D sound environment that simulates real-time spatial perception changes based on user interaction.
- Built a stable audio graph to handle complex panning and spatialization parameters.
- Synchronized high-frequency visual UI updates with audio state changes to eliminate latency and glitches.

DerbyOS Web | Vanilla JavaScript, HTML/CSS

- Architected a **Monte Carlo**—style race simulation using stochastic logic to manage randomized outcomes and betting flows.
- Built a custom state-management system without external frameworks to maintain simulation correctness and UI responsiveness.

TECHNICAL SKILLS

- Languages:** Python, C++, JavaScript (ES6+), SQL, LaTeX.
- Frameworks/Tools:** React, Django, FastAPI, MapLibre, Web Audio API, Git.
- Libraries:** NumPy, Qibo, Wolfram Alpha API.
- Core Competencies:** Quantum Simulation, Spatial Data, Monte Carlo Simulations, Conflict Resolution.

EDUCATION

- Simon Fraser University (SFU)** | *Bachelor of Arts in Cognitive Computing science* (Second Year).
- HarvardX (CS50X)** | *Online Computer Science Course*.
- Columbia College** | *University Transfer Program*.
- High School Diploma** | International Institution in Kenya. (Gulab Lochab Academy)

PROFESSIONAL EXPERIENCE

PC Express Lead (Superstore) | Richmond/Metrotown | [May 2023 – April 2025]

- Served as the **Weekend Lead** for 2 years, managing operations and guiding 3–4 employees in a high-volume retail environment.
- Collaborated with department managers to achieve service standards and departmental goals.
- Handled complex customer transactions and substitutions using advanced conflict resolution skills.

Sales Associate | JD Sports / Japara Limited | [July 2021 – Present]

- Assisted in opening the flagship **JD Sports** Robson Street location.
- Managed inventory systems, merchandise organization, and daily sales targets.

- Analyzed customer behavior to optimize stock ordering for high-interest automotive parts.
- Addressed customer complaints and provided data-driven feedback to the managing director regarding satisfaction.

REFERENCES Available upon request.
